

Instruction: Delete any examples in grey. Items that are not relevant should be removed as they are not part of the program. Spelling corrections are not required.									
Section A: Data Source Selection					Version				
Data source		Population Subset			2025q2				
Optum		Clinformatics Data Mart (CDM)							
Section B: Define Study Time Periods									
Study Time Period		Label		Definition		Additional Notes		Date Modified (Initials + Date)	
STUDYPD		Full study period		01Jul/2015 - 30June2025 (inclusive)		Full study time period		OK19012026	
ID_PD		Patient identification period		01Jan2016 - 30June2025 (inclusive)		Allows for 6mos lookback and no required minimum length of follow-up. ID_PD ends at the earliest of death or the end of the study period.		OK19012026	
INDEX		Index date		Date of the patient's earliest qualifying multiple myeloma diagnosis date		Patients are required to have at least 1 inpatient cancer visit for multiple myeloma (ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or at least 2 outpatient visits in any position on claim, on separate days within 90 days, during identification period (Patients with 2 claims 30 or 60 days apart will also be flagged for consideration.)		OK19012026	
						The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42 "MM dx code".			
						Flags should also be made to distinguish which claims are non-diagnostic claims. A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis (see tab 49 "Diagnostic").			
Baseline		Baseline period		The baseline period is fixed and is the six months before the index date.		Patients must have six months of CE with medical or pharmacy benefits before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026	
FU_PD		Follow up period		The follow up period is fixed and is from the index date till the end of the patient identification period (inclusive).		Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled; consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30June2025).		OK19012026	
						Note: Patients with ≥3 months of CE with medical or pharmacy benefits from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have have an assigned date			

1.Cover

2.QC Review

3.Data Prep

4.StudyPop

5A.MMA_MED

5B.MAP_MED

6.LOT1_BASE

7.SCT

10.LOT1_BASE_END OK

40.CL MMA ROLLUP

41.CL MMA CODELIST

41a.MMA NDC CODELIST

41b.MMA

A	B	C	D	E	F	G
		Approach 1: Use TOS_CD if conditions are not restricted on inpatient services associated with a hospitalization which requires an admission and discharge date. 1. Extract the records from T_Medical and T_CONFINEMENT datasets Inpatient -> POS IN (21, 51, 61) or TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF INPVS, FAC_IP SNF) Outpatient -> none of the above Approach 2: For anything related to services happening during a hospitalization with room and board, apply the condition on the CONF_ID because the majority of the work we do includes both diagnoses and medical services e.g. procedures, and when we talk about inpatient utilization, it is usually in the context of an admission and discharge. 1. Inpatient records should be restricted to cases where CONF_ID is not NULL, from the T_CONFINEMENT table, where this has associated admission and discharge dates. 2. The confinement table has summary information, so if users are pulling medical claims for specific inclusion, user should match on the CONF_ID variable, where it is not NULL in the confinement table. All other records without a CONF_ID or where CONF_ID is NULL should be considered non-inpatient.				
5	Identifying inpatient and non-inpatient claims		Approach. We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum CDM business rules guidance Approach 1 (POSTOS) Inpatient if POS IN (21, 51, 61) OR TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF INPVS, FAC_IP SNF) Approach 2 (CONF_ID) Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both ADMIT_DATE and DISCH_DATE present Non-inpatient: All records not meeting either approach above A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters	Rationale. Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data Total patients (Step 0): 94951 Any inpatient event: 43439 (45.2%) Via POSTOS (Apr 1): 42618 (44.3%) Via CONF_ID (Apr 2): 40357 (42.0%) Both approaches: 39536 (41.1%) POSTOS only: 3082 (3.2%) CONF_ID only: 821 (0.9%) Outpatient only: 52757 (54.8%)		OK19012026
6	Defining maintenance periods	Defining mtk periods is quite complex, especially with the elimination of the 90 day gap rule originally proposed by Optum. Without allowable (90d) gaps, for which mtk periods relied on for defining starts and stops, and due to the messy nature of claims and prescriber behavior, it is very difficult to create an overarching mtk rule.	For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapies/therapy combinations	Reporting mtk is not of importance/relevance to upcoming studies planned in Multiple Myeloma Flags will enable users to define mtk themselves in the future if so needed		
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